

Create a complete academic project report for the project titled:

“ComicCraftAI – AI-Powered Comic Story Creator”

The report should be professionally formatted and suitable for an academic AI/ML and Generative AI project submission.

Include the following sections in order:

1. Cover Page
2. Certificate
3. Declaration
4. Acknowledgement
5. Abstract
6. Table of Contents
7. List of Figures
8. List of Tables

CHAPTER 1 – INTRODUCTION

Explain the background of generative AI and AI-assisted creative content generation. Introduce ComicCraftAI and explain its purpose.

CHAPTER 2 – PROBLEM STATEMENT

Explain the difficulty users face when creating comics manually and define the problem addressed by ComicCraftAI.

CHAPTER 3 – OBJECTIVES

Explain the objectives of the project, including story generation, dialogue generation, image generation, comic layout, preview, and PDF export.

CHAPTER 4 – EXISTING SYSTEM

Describe traditional comic creation and its limitations.

CHAPTER 5 – PROPOSED SYSTEM

Describe ComicCraftAI as an AI-powered alternative that combines story generation, image generation, layout processing, and PDF generation.

CHAPTER 6 – REQUIREMENT ANALYSIS

Include functional requirements, non-functional requirements, software requirements, and hardware requirements.

CHAPTER 7 – SYSTEM DESIGN

Explain the system architecture, data flow diagram, module design, and user interface design.

CHAPTER 8 – TECHNOLOGY STACK

Explain the technologies used:

Python
FastAPI
HTML
CSS
Jinja2
Google Gemini API
Hugging Face Inference API
FPDF
Git
GitHub
Render

CHAPTER 9 – IMPLEMENTATION

Explain the implementation of:

FastAPI application
User input form
Comic outline generation
Story generation
Image generation
Comic layout
PDF generation
Comic preview

Explain the purpose of each major source code module.

CHAPTER 10 – TESTING

Explain the testing strategy and include test cases for home page loading, user input, AI generation, image generation, comic preview, PDF generation, and deployment.

CHAPTER 11 – RESULTS

Describe the expected project output: a five-panel illustrated comic containing scenes, captions, narration, dialogue, and generated images, together with PDF export.

CHAPTER 12 – DEPLOYMENT

Explain the deployment process using GitHub and Render. Explain environment variables and secure handling of API keys.

CHAPTER 13 – ADVANTAGES

Discuss the advantages of the system, including ease of use, reduced manual effort, AI-assisted creativity, automated image generation, and PDF export.

CHAPTER 14 – LIMITATIONS

Discuss realistic limitations such as dependence on external AI services, generation time, API availability, image consistency, and model limitations.

CHAPTER 15 – FUTURE ENHANCEMENTS

Discuss possible improvements such as:

- More comic panels
- Character consistency
- Multiple characters
- Comic editing
- Multiple layouts
- Voice narration
- User accounts
- Mobile optimization
- Additional art styles

CHAPTER 16 – CONCLUSION

Provide a conclusion describing how ComicCraftAI combines generative AI and web technologies to help users transform story ideas into illustrated comics.

REFERENCES

Include references to the official documentation of the technologies and APIs used in the project.

Use a formal academic writing style. Do not invent experimental results, performance measurements, user statistics, accuracy values, or testing results that were not actually obtained. Use placeholders where project-specific information is unavailable.